Phenomenological T_m models — formulas

Shared two-state core

$$T_m(^\circ\mathrm{C}) = \frac{1000\sum_k g_k \Delta H_{NN,k}}{\Delta S_{\mathrm{eff}}} - 273.15, \qquad g_k = e^{-\bar{d}_k/\lambda}$$

$\bar{d}_k$ = mean distance of bond k to the others; every ΔS_{eff} also includes $+R\ln(C_T/2) \approx -39.6$ cal/mol·K. $N =$ # bonds (2 per site).

Family A — occupancy-weighted | Family B — elastic / mechanical-coupling

A1 mean	$\Delta S_{\mathrm{eff}} = \sum_k g_k(\Delta S_{NN,k} + \Delta S_{Bond}) + \mathrm{base}$
A2 N-scaled ΔS_{Bond} (best)	$\Delta S_{\mathrm{eff}} = \sum_k g_k \Delta S_{NN,k} + N\,\Delta S_{Bond} + \mathrm{base}$
A3 g on ΔH only	$\Delta S_{\mathrm{eff}} = \sum_k \Delta S_{NN,k} + N\,\Delta S_{Bond} + \mathrm{base}$ (g applied only in ΔH)
A4 nearest-neighbor g	same as A1 with $g_k = e^{-d_{\min,k}/\lambda}$
A5 per-type ΔS_{Bond}	$\Delta S_{\mathrm{eff}} = \sum_k g_k(\Delta S_{NN,k} + \Delta S^{\mathrm{type}}_{Bond}) + \mathrm{base}$
A6 split λ_H/λ_S	ΔH uses $g(\lambda_H)$, ΔS uses $g(\lambda_S)$
A7 occ. + rotational	A2 $- k_{rot} R\ln(1 + \sum_k (z_k - \bar{z})^2)$
B1 g=1 (no weighting)	$\Delta S_{\mathrm{eff}} = \sum_k \Delta S_{NN,k} + N\,\Delta S_{Bond} + \mathrm{base}$ $(g_k \equiv 1)$
B2 g=1 + rotational	B1 $- k_{rot} R\ln(1 + \sum_k (z_k - \bar{z})^2)$
B3 g-weighted + WLC tethering	$\Delta S_{\mathrm{eff}} = \sum_k g_k(\Delta S_{NN,k} + \Delta S^{\mathrm{type}}_{teth}) + \mathrm{base}$
B4 WLC tethering only	$\Delta S_{\mathrm{eff}} = \sum_k (\Delta S_{NN,k} + \Delta S^{\mathrm{type}}_{teth}) + \mathrm{base}$

Mechanical terms

Rotational (elastic Green's fn): $\Delta S_{rot} = -k_{rot} R\ln(1 + \sum_k (z_k - \bar{z})^2)$

WLC tethering (derived): $\Delta S^{\mathrm{type}}_{teth} = R\ln\frac{V_2}{V_1^2 C_0}$, $V = \frac{4}{3}\pi\langle r^2\rangle^{3/2}$ (WLC, ℓ_p=1 nm, 0.63 nm/nt)

6nt is asymmetric (8 nt / 7 nt strands): $\Delta S^{6}_{teth} = R\ln\frac{V(1.5)}{V(8)\,V(7)\,C_0}$

Performance (10-fold CV, n = 235 interfaces)

Model	cv RMS (°C)	cv R^2	cv ρ	fit R^2	fitted params
A1 mean (g-weighted ΔS_Bond)	2.084	0.785	0.876	0.793	[53.15 -10.07 16.05]
A2 N-scaled ΔS_Bond	1.984	0.805	0.884	0.811	[139.82 -8.13 5.24]
A3 g on ΔH only	2.052	0.792	0.886	0.795	[1000. -7.27 4.23]
A4 nearest-neighbor g	2.561	0.676	0.748	0.679	[1000. -10.6 5.93]
A5 per-type ΔS_Bond	2.044	0.793	0.873	0.801	[57.54 -9.94 -9.19 11.82]
A6 split λ_H/λ_S	2.044	0.793	0.878	0.804	[484. 754.03 -8.1 5.52]
A7 occ.+rotational	2.001	0.802	0.881	0.813	[113.41 -7.98 -5.34 -0.86]
B1 g=1 (no weighting)	2.561	0.676	0.748	0.679	[-10.6 5.8]
B2 g=1 + rotational	2.199	0.761	0.86	0.775	[-9.16 48.08 3.49]
B3 g-weighted + WLC tethering	4.047	0.19	0.547	0.201	[73.82 48.28]
B4 WLC tethering only	5.888	-0.715	0.636	-0.699	[1000. 77.11]